

BTEC Level 3 Applied Science • Unit 1 Chemistry • Answer sheet

Atomic Structure — Answers

Mark scheme for the companion worksheet. Accept equivalent correct wording. These are original JDSscience items, not official Pearson questions.

Numbering matches the student worksheet exactly. Award equivalent scientifically correct wording. Show units on every calculated quantity.

A

1. Proton +1, mass 1. Neutron 0, mass 1. Electron -1, mass $\approx 1/1836$.
2. Z = number of protons. A = protons + neutrons.
3. Same proton number / element; different neutron number / mass number.

B

4. 19 p, 20 n, 19 e.
5. Number of protons equals number of electrons, so charges cancel.

C

6. Mg: 12p 12n 12e. Mg^{2+} : 12p 12n 10e. F: 9p 10n 9e. F^- : 9p 10n 10e.

7. $A_r = (60 \times 69 + 40 \times 71) / 100 = 69.8$

Equation	$A_r = (\Sigma \text{ percent} \times \text{mass number}) / 100$
Substitution	$A_r = (60.0 \times 69 + 40.0 \times 71) / 100$
Working	$(4140 + 2840) / 100 = 6980 / 100$
Answer + unit	69.8 (no unit)

Marking guidance: 1 mark substitution with both isotopes; 1 mark 69.8.

8. Chlorine is a mixture of ^{35}Cl and ^{37}Cl ; A_r is a weighted mean.

D / E

9. Isotopes must have the same proton number. C has 6 protons; N has 7.

10. $(75 \times 63 + 25 \times 65) / 100 = 63.5$

Equation	$A_r = (\Sigma \text{ percent} \times \text{mass number}) / 100$
Substitution	$A_r = (75 \times 63 + 25 \times 65) / 100$
Working	$(4725 + 1625) / 100 = 6350 / 100$
Answer + unit	63.5 (no unit)

Marking guidance: 1 mark substitution; 1 mark 63.5.

11. The atom gains two electrons. Proton number unchanged. Electrons exceed protons by 2, so charge is 2⁻. [3]

12. $A_r = (90.5 \times 20 + 0.27 \times 21 + 9.25 \times 22) / 100 = 20.2$ (3 s.f.). Ignoring ^{21}Ne changes only the third significant figure.

Equation	$A_r = (\Sigma \text{abundance} \times \text{mass number}) / 100$
Substitution	$A_r = (90.5 \times 20 + 0.27 \times 21 + 9.25 \times 22) / 100$
Working	$(1810 + 5.67 + 203.5) / 100 = 20.2$ (3 s.f.)
Answer + unit	20.2 (3 s.f.)

Marking guidance: 1 mark full substitution; 1 mark 20.2; comment that ignoring ^{21}Ne barely changes 3 s.f.

13. A = 23 top left, Z = 11 bottom left, 11 p, 12 n. Na^+ has 10 electrons.

Nuclide notation • sodium-23

$$\begin{matrix} 23 \\ 11 \end{matrix} \text{Na}$$

Mass number A = 23
protons + neutrons

Atomic number Z = 11
protons — identifies the element

Chemical symbol
Na is still sodium if the charge changes

Neutrons = A - Z = 23 - 11 = 12

Neutral atom: electrons = Z = 11

Na^+ ion: still 11 protons; electrons = 10. The nucleus does not change.